Olist E-Commerce Analytics

Complete Technical Documentation, Advanced SQL Diagnostics, Data Modeling & Business Intelligence

Project Type: Graduation Capstone

Domain: Brazilian Marketplace B2B2C

Timeline: 2016 – 2018 (99,441 Orders)

Tech Stack: MySQL, Power BI, Power Pivot, DAX

R$ 15.42M	96,096	3.10%	4.10 ★
GROSS DELIVERED REVENUE	UNIQUE CUSTOMERS	REPEAT CUSTOMER RATE	AVG. REVIEW RATING

1. PROJECT BACKGROUND, BUSINESS UNDERSTANDING & PROBLEM STATEMENT

Olist is a Brazilian technology startup founded in 2015 that operates as a marketplace integrator (B2B2C business model). It connects thousands of small and medium-sized physical and digital merchants across Brazil to major e-commerce platforms—including Amazon Brazil, Mercado Livre, B2W, and Magazine Luiza—through a single management contract and unified dashboard. Olist handles product catalog distribution, localized logistics fulfillment, and payment operations on merchants' behalf.

Core Analytical Objective: By examining the full transactional dataset covering 99,441 orders placed between September 2016 and October 2018, this capstone project investigates the operational and commercial mechanisms driving the platform. The analysis seeks to resolve critical business questions:

- What are the core drivers of marketplace revenue growth, and where is geographic and category concentration creating vulnerability?
- Why is customer retention critically low (3.1%), and what is the commercial value of recurring buyers?
- What operational factors drive customer dissatisfaction, and does delivery speed outweigh pricing in determining review ratings?
- Which seller cohorts and payment channels introduce systemic operational risk (cancellations and delivery delays)?

2. COMPREHENSIVE DATA ARCHITECTURE & RELATIONAL DATA MODEL

The database was structured into a high-performance **Star** / **Snowflake Schema** in Power Pivot and Power BI, consisting of 4 core Fact tables and 5 Dimension lookup tables. Every foreign key relationship operates on a verified 1-to-Many grain with zero orphan records.

Table Name	Type	Row Count	Primary Key	Foreign Keys & Relationships
`Fact_Orders`	Fact	99,441	`order_id`	`customer_id` → `Dim_Customers` `order_purchase_timestamp` → `Dim_Calendar`

Table Name	Type	Row Count	Primary Key	Foreign Keys & Relationships

`Fact_OrderItems`	Fact	112,650	`order_id` + `order_item_id`	`order_id` → `Fact_Orders`
				`product_id` → `Dim_Products`
				`seller_id` → `Dim_Sellers`

`Fact_Payments`	Fact	103,886	`order_id` + `payment_sequential`	`order_id` → `Fact_Orders`
`Fact_Reviews`	Fact	98,410	`review_id`	`order_id` → `Fact_Orders`
`Dim_Customers`	Dimension	99,441	`customer_id`	`customer_zip_code_prefix` → `Dim_Geolocation`
`Dim_Products`	Dimension	32,951	`product_id`	`product_category_name` (lookup translation)
`Dim_Sellers`	Dimension	3,095	`seller_id`	`seller_zip_code_prefix` → `Dim_Geolocation`
`Dim_Geolocation`	Dimension	19,015	`geolocation_zip_code_prefix`	Lookup Key for Customers & Sellers
`Dim_Calendar`	Dimension	800	`Date`	`Date` → `Fact_Orders[order_purchase_timestamp]`

Data Scope & Known Boundaries: Customer gender and Product Cost of Goods Sold (COGS) are not collected in Olist's raw logs. Consequently, commercial metrics represent *Gross Merchandise Value (GMV) / Gross Revenue* rather than net profit margins.

3. END-TO-END ETL, CLEANING & DATA MODELING ENGINEERING

Transforming the raw CSV files into an enterprise-grade reporting model required solving critical data quality challenges:

3.1. Geolocation Master Deduplication

The raw `olist_geolocation_dataset.csv` contained 1,000,163 rows with multiple GPS pings recorded for the same 5-digit zip code prefix. Direct joining created severe Cartesian product explosion.

Solution: The dataset was grouped by `geolocation_zip_code_prefix` in SQL and Power Query, averaging latitude and longitude coordinates and taking the statistical mode for city/state names, reducing the table to 19,015 distinct, clean centroid records.

3.2. Customer Identity Resolution

Olist generates a new `customer_id` for every transaction. Counting `customer_id` creates an artificial 100% one-time buyer rate.

Solution: The actual human entity is tracked via `customer_unique_id`. Grouping across `customer_unique_id` revealed that 96,096 unique individuals placed the 99,441 orders, accurately exposing the 3.10% repeat purchase rate.

3.3. Date Range Reliability & Edge-Period Filtering

The dataset spans Sep 2016 to Oct 2018. However, boundary months have sparse records (Sep 2016 = 4 orders; Nov 2016 = 0; Dec 2016 = 1; Sep 2018 = 16; Oct 2018 = 4). Including these raw boundaries in time-series visuals generates misleading artificial business collapses.

Solution: Trend analyses and monthly run-rates were programmatically bounded to the fully active operating window (**January 1, 2017 to August 31, 2018**).

3.4. Category Normalization & Missing Values

Two active product categories (`pc_gamer` and `portateis_cozinha_e_preparadores_de_alimentos`) were missing from the translation lookup table, while 610 products lacked category assignments.

Solution: Added explicit dictionary translations and classified unassigned items into an *"Unknown/Other"* category to ensure zero line items or revenues were dropped.

4. COMPREHENSIVE SQL ANALYSIS (26 ADVANCED DIAGNOSTIC QUERIES)

To perform comprehensive exploratory data analysis and answer all operational and commercial questions, 26 SQL queries were executed in MySQL. Below are the key queries, categorized by business domain:

4.1. Revenue, Order Volume & Category Performance

Analysis 1: Total Delivered Revenue

```
SELECT
    ROUND(SUM(oi.price + oi.freight_value), 2) AS total_delivered_revenue
FROM Fact_OrderItems oi
JOIN Fact_Orders o ON oi.order_id = o.order_id
WHERE o.order_status = 'delivered';
-- Result: R$ 15,422,461.77
```

Analysis 2: Monthly Revenue Run-Rate Trend (Clean Operating Window)

```
SELECT
    YEAR(o.order_purchase_timestamp) AS order_year,
    MONTH(o.order_purchase_timestamp) AS order_month,
    ROUND(SUM(oi.price + oi.freight_value), 2) AS monthly_revenue,
    COUNT(DISTINCT o.order_id) AS total_orders
FROM Fact_Orders o
JOIN Fact_OrderItems oi ON o.order_id = oi.order_id
WHERE o.order_status = 'delivered'
  AND o.order_purchase_timestamp >= '2017-01-01'
  AND o.order_purchase_timestamp < '2018-09-01'
```

```
GROUP BY 1, 2
ORDER BY 1, 2;
```

Analysis 3: Revenue Contribution by Product Category

```
SELECT
    COALESCE(p.product_category_name_english, 'Unknown') AS product_category,
    COUNT(DISTINCT oi.order_id) AS total_orders,
    COUNT(*) AS items_sold,
    ROUND(SUM(oi.price + oi.freight_value), 2) AS category_revenue,
    ROUND(SUM(oi.price + oi.freight_value) * 100.0 /
          (SELECT SUM(price + freight_value) FROM Fact_OrderItems oi2
           JOIN Fact_Orders o2 ON oi2.order_id = o2.order_id WHERE o2.order_status = 'delivered'), 2)
AS revenue_share_pct
FROM Fact_OrderItems oi
JOIN Fact_Orders o ON oi.order_id = o.order_id
JOIN Dim_Products p ON oi.product_id = p.product_id
WHERE o.order_status = 'delivered'
GROUP BY 1
ORDER BY category_revenue DESC
LIMIT 10;
-- Top: Health & Beauty (R$ 1.41M), Watches & Gifts (R$ 1.39M), Bed Bath Table (R$ 1.37M)
```

4.2. Customer Segmentation & Geographic Distribution

Analysis 4: Customer Retention & Repeat Purchase Segmentation (Window Functions)

```
WITH CustomerOrderStats AS (
    SELECT
        c.customer_unique_id,
        COUNT(DISTINCT o.order_id) AS total_orders,
        SUM(oi.price + oi.freight_value) AS customer_monetary_val
    FROM Dim_Customers c
    JOIN Fact_Orders o ON c.customer_id = o.customer_id
    JOIN Fact_OrderItems oi ON o.order_id = oi.order_id
    WHERE o.order_status = 'delivered'
    GROUP BY c.customer_unique_id
)
SELECT
    CASE
        WHEN total_orders = 1 THEN 'One-Time Buyer (1 Order)'
        WHEN total_orders BETWEEN 2 AND 3 THEN 'Repeat Buyer (2-3 Orders)'
        ELSE 'VIP / Power Buyer (4+ Orders)'
    END AS customer_segment,
    COUNT(*) AS total_customers,
    ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 2) AS pct_customer_base,
    ROUND(AVG(customer_monetary_val), 2) AS avg_revenue_per_customer,
    ROUND(SUM(customer_monetary_val), 2) AS total_segment_revenue
FROM CustomerOrderStats
GROUP BY 1
ORDER BY total_customers DESC;
-- Insight: One-Time = 96.9%, Repeat = 3.0%, VIP = 0.1% (VIPs spend 4.8x higher per customer)
```

Analysis 5: Regional Revenue Concentration by State

```
SELECT
    c.customer_state,
    COUNT(DISTINCT c.customer_unique_id) AS unique_customers,
    COUNT(DISTINCT o.order_id) AS total_orders,
```

```
    ROUND(SUM(oi.price + oi.freight_value), 2) AS state_revenue,
    ROUND(SUM(oi.price + oi.freight_value) * 100.0 /
          (SELECT SUM(price + freight_value) FROM Fact_OrderItems oi2
           JOIN Fact_Orders o2 ON oi2.order_id = o2.order_id WHERE o2.order_status = 'delivered'), 2)
AS state_rev_pct
FROM Fact_Orders o
JOIN Fact_OrderItems oi ON o.order_id = oi.order_id
JOIN Dim_Customers c ON o.customer_id = c.customer_id
WHERE o.order_status = 'delivered'
GROUP BY c.customer_state
ORDER BY state_revenue DESC
LIMIT 5;
-- Top 5 States generate 73.2% of revenue: SP (37.4%), RJ (13.2%), MG (12.1%), RS (5.6%), PR (4.9%)
```

4.3. Logistics, Delivery Performance & Satisfaction Drivers

Analysis 6: On-Time vs. Late Delivery Rate & Punctuality KPIs

```
SELECT
    CASE
        WHEN order_delivered_customer_date <= order_estimated_delivery_date THEN 'On-Time'
        ELSE 'Late Delivered'
    END AS delivery_compliance,
    COUNT(*) AS delivered_orders,
    ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 2) AS compliance_rate,
    ROUND(AVG(DATEDIFF(order_delivered_customer_date, order_purchase_timestamp)), 2) AS
avg_transit_days,
    ROUND(AVG(CASE WHEN order_delivered_customer_date > order_estimated_delivery_date
                   THEN DATEDIFF(order_delivered_customer_date, order_estimated_delivery_date)
                   ELSE NULL END), 2) AS avg_days_delayed
FROM Fact_Orders
WHERE order_status = 'delivered'
  AND order_delivered_customer_date IS NOT NULL
  AND order_estimated_delivery_date IS NOT NULL
GROUP BY 1;
-- Result: On-Time = 93.34% (12.05 avg days), Late = 6.66% (average 10.5 days past promised date)
```

Analysis 7: Customer Rating Collapsing Effect (On-Time vs. Late Orders)

```
SELECT
    CASE
        WHEN o.order_delivered_customer_date <= o.order_estimated_delivery_date THEN 'On-Time
Delivery'
        WHEN o.order_delivered_customer_date > o.order_estimated_delivery_date THEN 'Late Delivery'
        ELSE 'Undelivered / Unknown'
    END AS fulfillment_status,
    COUNT(r.review_score) AS review_count,
    ROUND(AVG(r.review_score), 2) AS average_review_score,
    ROUND(SUM(CASE WHEN r.review_score = 5 THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS
five_star_pct,
    ROUND(SUM(CASE WHEN r.review_score <= 2 THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS
dissatisfied_pct
FROM Fact_Orders o
JOIN Fact_Reviews r ON o.order_id = r.order_id
WHERE o.order_status = 'delivered'
GROUP BY 1;
-- Insight: On-time orders average 4.29★ (10.1% low rating). Late orders collapse to 2.27★ (64.8% low
rating)
```

4.4. Payment Methods & Operational Cancellation Risk

Analysis 8: Payment Gateway Mix & Cancellation Rate Disparity

```
SELECT
    p.payment_type,
    COUNT(DISTINCT p.order_id) AS total_orders_attempted,
    ROUND(SUM(p.payment_value), 2) AS total_payment_value,
    SUM(CASE WHEN o.order_status = 'canceled' THEN 1 ELSE 0 END) AS canceled_orders,
    ROUND(SUM(CASE WHEN o.order_status = 'canceled' THEN 1 ELSE 0 END) * 100.0 / COUNT(DISTINCT
p.order_id), 2) AS cancellation_rate_pct
FROM Fact_Payments p
JOIN Fact_Orders o ON p.order_id = o.order_id
GROUP BY p.payment_type
ORDER BY cancellation_rate_pct DESC;
-- Insight: Voucher cancellation is 3.21% (approx. 6x higher than Credit Cards at 0.57% and Boletos
at 0.48%)
```

5. DAX FORMULAS & ENTERPRISE MEASURE REPOSITORY

All business logic in Power BI was standardized into DAX measures housed within a dedicated `_Measures` table:

```
/* 1. COMMERCIAL & SALES MEASURES */
Total Revenue =
SUMX (
    Fact_OrderItems,
    Fact_OrderItems[price] + Fact_OrderItems[freight_value]
)

Total Orders = DISTINCTCOUNT ( Fact_Orders[order_id] )

Average Order Value (AOV) = DIVIDE ( [Total Revenue], [Total Orders], 0 )

Revenue per Customer = DIVIDE ( [Total Revenue], [Total Customers], 0 )

/* 2. CUSTOMER RETENTION & SEGMENTATION MEASURES */
Total Customers = DISTINCTCOUNT ( Dim_Customers[customer_unique_id] )

Repeat Customers =
COUNTROWS (
    FILTER (
        VALUES ( Dim_Customers[customer_unique_id] ),
        CALCULATE ( DISTINCTCOUNT ( Fact_Orders[order_id] ) ) > 1
    )
)

Repeat Customer Rate = DIVIDE ( [Repeat Customers], [Total Customers], 0 )

/* 3. LOGISTICS & FULFILLMENT MEASURES */
Average Delivery Days =
AVERAGEX (
    FILTER (
        Fact_Orders,
        NOT ISBLANK ( Fact_Orders[order_delivered_customer_date] )
            && NOT ISBLANK ( Fact_Orders[order_purchase_timestamp] )
    ),
    DATEDIFF (
        Fact_Orders[order_purchase_timestamp],
        Fact_Orders[order_delivered_customer_date],
        DAY
```

```
    )
)

Late Delivered Orders =
CALCULATE (
    [Total Orders],
    FILTER (
        Fact_Orders,
        NOT ISBLANK ( Fact_Orders[order_delivered_customer_date] )
            && NOT ISBLANK ( Fact_Orders[order_estimated_delivery_date] )
            && Fact_Orders[order_delivered_customer_date] >
Fact_Orders[order_estimated_delivery_date]
    )
)

On-Time Delivered Orders =
CALCULATE (
    [Total Orders],
    FILTER (
        Fact_Orders,
        NOT ISBLANK ( Fact_Orders[order_delivered_customer_date] )
            && NOT ISBLANK ( Fact_Orders[order_estimated_delivery_date] )
            && Fact_Orders[order_delivered_customer_date] <=
Fact_Orders[order_estimated_delivery_date]
    )
)

Late Delivery Rate = DIVIDE ( [Late Delivered Orders], [Total Orders], 0 )
On-Time Delivery Rate = DIVIDE ( [On-Time Delivered Orders], [Total Orders], 0 )

/* 4. COMPOSITE SELLER HEALTH SCORE (0 - 100 WEIGHTED INDEX) */
Seller Composite Score =
VAR ReviewWeight = DIVIDE ( [Seller Avg Review Score], 5, 0 ) * 40
VAR OnTimeWeight = [Seller On-Time Rate] * 30
VAR SalesWeight = [Seller Sales Normalized] * 20
VAR CancellationWeight = ( 1 - [Seller Cancellation Rate] ) * 10
RETURN
    ReviewWeight + OnTimeWeight + SalesWeight + CancellationWeight

Sellers in Danger =
COUNTROWS (
    FILTER (
        VALUES ( Dim_Sellers[seller_id] ),
        [Seller Composite Score] < 50 && [Total Orders] >= 5
    )
)
```

6. POWER BI INTERACTIVE DASHBOARD ARCHITECTURE

The report interface consolidates all analytical findings into an executive 3-page storytelling flow:

Dashboard Page	Strategic Question	Key Visuals & Interactive Elements	Business Value & Narrative
Page 1: Executive Overview	How is the marketplace expanding, and where is growth concentrated?	• KPI Metric Cards (R$ 15.4M Revenue, 99.4K Orders, R$ 137.7 AOV) • Monthly Revenue Run-Rate (Line Chart with Period Slicers) • Geographic Revenue Distribution (Filled Shape Map of Brazil) • Order Fulfillment Funnel (Delivered vs. Shipped vs. Cancelled)	Demonstrates steady business acceleration through mid-2018 while exposing severe structural revenue dependence on São Paulo (37.4%) and Southeast Brazil.
Page 2: Delivery & Customer Satisfaction	Where are logistics bottlenecks destroying customer goodwill?	• Combined Dual-Axis Trend: Average Delivery Days vs. Review Score • Late Delivery Rate by Product Category (Horizontal Bar Chart) • Customer Review Rating Breakdown (1★ to 5★ Donut Chart) • Freight Ratio Benchmark: Freight Cost as % of Product Price	Validates that delivery duration directly dictates ratings. Late deliveries incur an immediate 2.02★ penalty. Highlights heavy goods categories suffering freight drag.
Page 3: Marketplace & Seller Risk	Which merchant segments and payment pipelines present operational risk?	• Dynamic Customer Segmentation: One-time vs. Repeat vs. VIP • Revenue per Customer by Segment (Clustered Column Chart) • Seller Risk Matrix: Seller Revenue vs. Composite Score (Scatter Plot) • Cancellation Rate by Payment Method (Clustered Bar Chart)	Isolates high-volume merchants operating with dangerously low fulfillment scores (<50) and pinpoints voucher payment gateway cancellation vulnerabilities (3.21%).

7. KEY DIAGNOSTIC INSIGHTS & STRATEGIC RECOMMENDATIONS

1. Prioritize Courier SLA Optimization Over Price Subsidies: Statistical correlation demonstrates that delivery delay is the primary driver of customer dissatisfaction ($r = -0.33$), whereas product price ($r = -0.005$) and freight value ($r = -0.03$) have virtually zero relationship with review scores. Lowering prices will not improve satisfaction if shipping is slow.

2. Deploy an Automated Late-Delivery Recovery Workflow: While late deliveries represent only 6.66%–8.10% of orders, they trigger a catastrophic rating collapse from 4.29★ to 2.27★ and average 10.5 days of delay. Olist must implement automated dispatch alerts and issue proactive credit vouchers *before* the customer complains.

3. Institute Customer Retention & Re-engagement Programs: With 96.9% of buyers ordering only once, Customer Acquisition Cost (CAC) heavily burdens unit economics. Given that VIP buyers generate 4.8× higher spend than one-time buyers, automated post-purchase cross-selling campaigns within 30 days are essential.

4. Expand Regional Logistics Hubs (North / Northeast): Because 73.2% of sales originate from 5 southeastern states, long-distance interstate transit to the North/Northeast causes elevated transit times (up to 28 days in RR and AP). Establishing regional fulfillment centers will unlock national growth.

5. Audit Voucher Gateway & Verification Workflow: Vouchers exhibit an alarming 3.21% cancellation rate (6× higher than Credit Cards at 0.57%). Engineering must audit coupon expiration timers, checkout validation logic, and automated refund re-issuance to eliminate checkout drop-off.